

Unequal Job Opportunities and the Measurement of Welfare Inequality

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France 2016 · EU-SILC priced through EUROMOD

[0.5 min] Say the one-sentence version of the paper and move.

"I estimate a labour-supply model in which each household faces its own distribution of available jobs, and then I ask how much of welfare inequality is about preferences and how much is about the environment people face."

Do not preview the answer yet; the next slide earns it. Twenty seconds.

Two people, the same job, different reasons

Person A

40 hours

same occupation

bottom wage quintile

chose it from a wide set

Person B

40 hours

same occupation

bottom wage quintile

had almost nothing else

Whatever the choice set cannot express,
the taste parameter absorbs.

[1.5 min] This is the motivating slide and it should be told, not read.

Two people make the same choice. One had a wide range of jobs open and picked this one; the other had almost nothing else available.

A model that puts both on a common budget set must book the entire difference as *taste* — and will then report that they are equally well off, or that the second one simply prefers what she has.

That is not a measurement error at the margin; it is the central identification problem in applied welfare analysis of labour supply.

Land the sentence: whatever the choice set cannot express, the taste parameter absorbs.

25-MIN: compress to 1.0 — one vignette, not two.

(A) same preferences, different opportunity environments

(B) same opportunity environment, different preferences

(C) job access / feasibility

(D) job access / feasibility

consumption c (schematic units)

leisure $l = \bar{L} - h$ (schematic units)

consumption c (schematic units)

leisure $l = \bar{L} - h$ (schematic units)

offer density g^h

weekly hours h

offer density g^h

weekly hours h

earning opportunities, by occupation

offer density $g^{h,c}(\cdot)$

hourly wage offer w

earning opportunities, by occupation

offer density $g^{h,c}(\cdot)$

hourly wage offer w

— indifference curve (attained) ● the chosen job — household 1 / 3
 attained welfare $\bar{W}$ (money metric) — household 2 / 4

— household 1 / 3
 — household 2 / 4

Schematic - No estimated quantity

Say explicitly what the figure is: **a schematic, drawn from stylised parameters, with no estimated quantity on any axis.**

Say what the opportunity object is and what it is not — it is a *density over latent job packages*, not a list and not a count.

Flag that slide 14 shows the same two-panel structure on two real households.

25-MIN: compress to 1.5 — panel (a) only.

2/22

Why random opportunities, not a common choice set

	common choice set	random opportunities
what varies across households	the budget only	the whole offer density
unexplained variation goes to	preferences	job access, earning opportunities, preferences
welfare comparison available	levels at a common set	levels at own set
what is identified	tastes	tastes <i>and</i> availability

Remove the occupation term and the effect does not vanish — it relocates.

tertiary-education loading moves by six standard errors

[1.5 min] Be fair to the alternative: a common-choice-set model is not wrong, it answers a different question, and it is cheaper.

The point is structural. If every household faces the same set, then differences in observed choices *must* load on preferences, and a welfare measure built on it inherits that.

Then give the paper’s own internal evidence rather than an assertion — this is the strongest positive-side argument in the paper, so do not rush it. When the occupation term is removed from the earning-opportunity block, the effect does not vanish: it **relocates**, with the tertiary-education loading moving by six standard errors and the female nonroutine-cognitive access coefficient moving with it.

The same variation gets booked to whichever channel the model leaves open. That is a channel-separation result, and it is exactly what a decomposition has to get right.

Slide 19 carries the empirical version of this argument; this slide states the structural point and forwards to it.

25-MIN: compress to 1.0 — the structural point and one line of the relocation result.

What the opportunity distribution is made of



estimated jointly with Box–Cox **preferences** in a single likelihood

The hours factor carries no household-specific coordinate.

[2.0 min] Name each factor and what varies with the household.

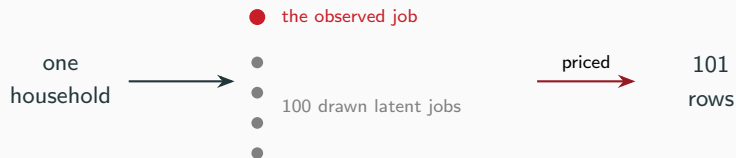
Access varies with the group-specific unemployment rate, region and urbanisation; occupation availability is sex-specific; the wage-offer location varies with education and experience and shifts by occupation.

Say the honest structural caveat here rather than burying it: **the hours factor carries no household-specific coordinate**, so household variation in the offer density lives in the participation margin, the occupation weights, and the wage location — and nowhere else. Saying that yourself, early, is worth more than being asked.

The decomposition later groups these as job access, earning opportunities, and household endowments and needs.

25-MIN: compress to 1.0 — the four boxes and the hours caveat, nothing else.

Data and the sampled-job estimator



EU-SILC / SRCV 2016 · 1,555 households · 101 alternatives each · 157,055
priced rows

conditional logit with the sampling-of-alternatives correction

[2.0 min] One slide, and deliberately no implementation detail — no check counts, no convergence protocol, no hash.

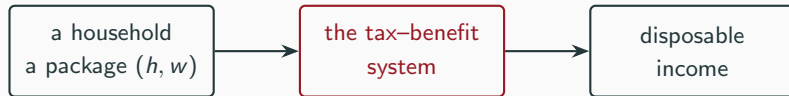
The job space is continuous in wage and hours, so the choice set is sampled rather than enumerated, and the sampling-of-alternatives correction is what makes the sampled-set likelihood consistent for the full-set parameters.

Say the one consequence the audience will otherwise trip on: because the observed row is inserted deterministically and the drawn rows carry a proposal correction, **within-set rank and top-k statistics are structurally degenerate here** — they are not reported, and their absence is a property of the estimator, not a weakness being hidden.

If asked about the technical layer, point to the appendix and move on.

25-MIN: MERGE with slide 7 (2.5 combined).

Pricing: what a job is worth



Consumption is computed, not surveyed.

The mapping is deterministic: the same household, hours and wage always return the same euro figure.

Nothing in the estimation simulates a tax-benefit outcome.

[2.0 min] This is the slide that makes the counterfactual legitimate, and it is worth two clean minutes.

EUROMOD is arithmetic: the same household, hours and wage always return the same euro figure. **Nothing in the estimation simulates a tax-benefit outcome.** All the randomness in the pipeline is in *which* points get evaluated, never in what a point is worth. That is what lets the model price a job the person did not take.

Mention take-up once — the minimum-income benefit is granted in full inside the engine and take-up is applied afterwards as a household trait — and say why: take-up is a property of the household, not of the batch it was priced in.

25-MIN: MERGE into slide 6.

Identification: what is pinned, what stays conditional

what the data pin	what stays conditional
the four availability factors	access <i>versus</i> capability
the preference block	persistent household heterogeneity
the statutory-week peak	

The decomposition is structural and model-conditional. It is not causal.

[2.0 min] Two conditional items, said plainly.

First, **the access margin does not separate what a person is capable of from what the market offers them** — the estimated access density may combine both, and this paper claims no ordering between the job-access and earning-opportunity channels in either direction, even though the point estimates order them and are printed.

Second, the model carries no persistent household-level term, and slide 9 is why.

Say the general disclaimer once, here: the decomposition is **structural and model-conditional, and it is not causal**.

25-MIN: KEEP (2.0) — and add one sentence of slide 9's lesson: *“three heterogeneity extensions were tested and all three failed recoverability, for three different reasons; the backup slide has the design rule.”*

Three heterogeneity extensions

extension	what defeats recovery
random leisure intercept	no leverage
working-alternative frailty	leverage, but no second contrast
wage-residual dependence	no leverage

Depth on an axis is not evidence; only depth on a profile is.

axis 13.56 → profile 0.6996 against 2.706

[2.0 min] This is the slide that turns an appendix of failures into a result, and in a methods-literate audience it is often the most interesting five minutes of the talk.

All three were specified in advance and assessed by synthetic-recovery diagnostics *before* any real-data estimation.

Two conditions must hold together for a persistent household-level term to be identified here. **Leverage:** the variable it loads on must have enough within-household spread across the choice set, measured against a scale-one Gumbel, to reorder a household’s own alternatives. The Box–Cox leisure transform has a within-set spread of 0.073 for men and 0.127 for women — that is the wall the random intercept and the wage-residual extension both hit.

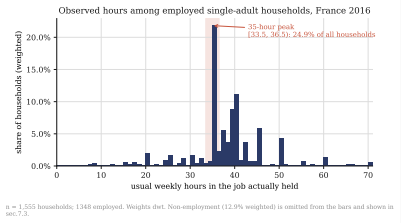
A second contrast: even with leverage, the term must move something the existing coordinates cannot absorb. The frailty term has four times the leverage and still fails, because a term that shifts every working alternative equally is a shift the employment intercept can simply re-fit, and with one choice per household there is no second margin to pin them apart.

Then the punchline: **depth on an axis is not evidence; only depth on a profile is.** Holding the rest of the model fixed gives that extension a likelihood-ratio depth of up to 13.56; re-optimising the rest gives 0.6996, against the 2.706 a boundary test needs.

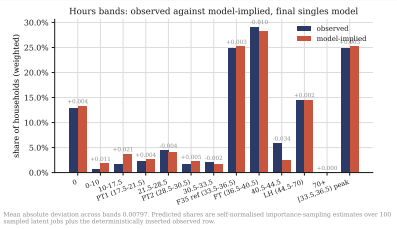
Close by saying what this does *not* say: it rejects three parameterisations on this design, not the existence of heterogeneity.

25-MIN: CUT to backup; fold one sentence into slide 8.

Hours, and the statutory week



observed hours



observed against model-implied

statutory band 24.85% observed · 25.20% model-implied · mean absolute error 0.0083 over 12 bins

[2.0 min] Observed 24.85 per cent in the statutory band against 25.20 model-implied; mean absolute error 0.0083 across 12 bins.

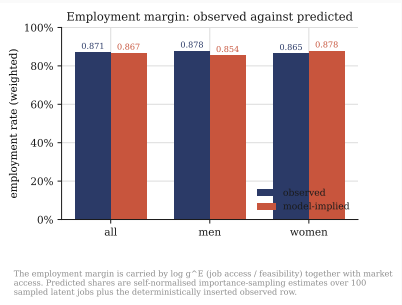
The statutory-week coordinate is the single largest one-coordinate improvement in the specification — a likelihood-ratio statistic of 861 on one degree of freedom.

Then the wording discipline, and use exactly this wording: it is an **institutionally motivated opportunity peak in the estimated offer density**. It is *not* an estimate of the causal effect of the 35-hour statute, and no counterfactual removing the statute is computed anywhere.

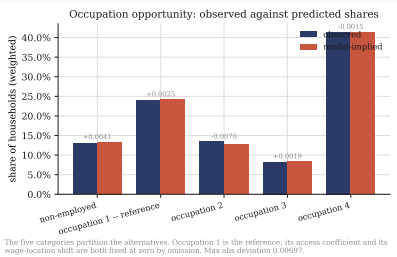
Concede the identified weak spot without being asked: the model under-predicts the 40.5–44.5 band by -0.034 — 3.4 points.

25-MIN: MERGE with slide 11 (2.0 combined).

Employment, occupation, and wages



employment



occupation shares

employment 0.8708 / 0.8668 · occupation largest deviation 0.0070 · median wage fit -0.86 men, +0.34 women (€/h)

[1.5 min] Employment 0.8708 observed against 0.8668 model-implied. Occupation: largest deviation across the 5 categories, including non-employment, is 0.0070 — seven-tenths of a percentage point.

Then be straightforward about the wage side, because a referee will find it anyway: the model **over-predicts the bottom offered-wage quintile by 5.3 points** and under-predicts the second, third and fourth.

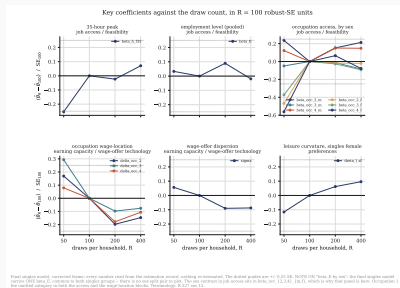
On selected wages the median fit is **-0.86 euros an hour for men and +0.34 for women** — the model slightly compresses the sex gap while matching the pooled median to under thirty cents (-0.29).

Say that a repair specified in advance was tried — an hours-conditioned wage location — and that it **failed**: it did not fix the misfit and was not promoted.

Occupation fit is shown *internally* here for a reason slide 13 explains.

25-MIN: MERGE into slide 10; keep employment, occupation and the wage misfit; drop the by-sex wage detail.

Does the answer depend on the number of drawn jobs?



key coordinates, in SE units

No. Stable from 50 to 400 drawn alternatives.

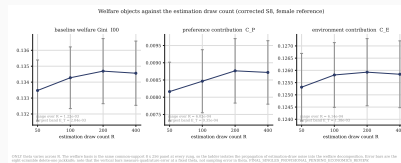
[1.0 min] Thirty seconds of content and a minute of credibility.

Every sign is constant and the nested order holds at every rung.

State the one exception rather than claiming uniformity: in the male-reference arm the preference contribution moves by 1.44 times its own band across the ladder — and it is also the object with the tightest band in the study, so the exceedance is as much about the band as the movement.

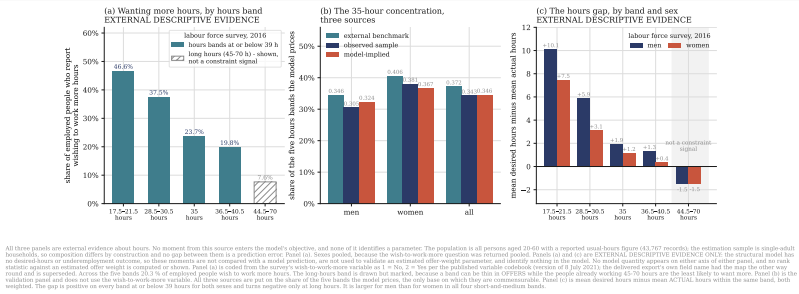
The consequence is a reporting rule: **the male-reference preference contribution is not quoted to more than two significant figures.**

25-MIN: CUT to backup.



the headline contributions

External validation



statutory band 37.2% external · 34.3% observed · 34.6% model-implied

[2.5 min] Frame it before showing it: **validation only — no moment from any external source enters the likelihood, and none of this identifies anything.**

Panel (b) is the one validation panel: the statutory band takes 37.2 per cent of workers in the external source, 34.3 in the observed sample and 34.6 model-implied, and the concentration is larger for women than men in all three (40.6 against 34.6 externally) — which the model reproduces **without any sex-specific hours coordinate**.

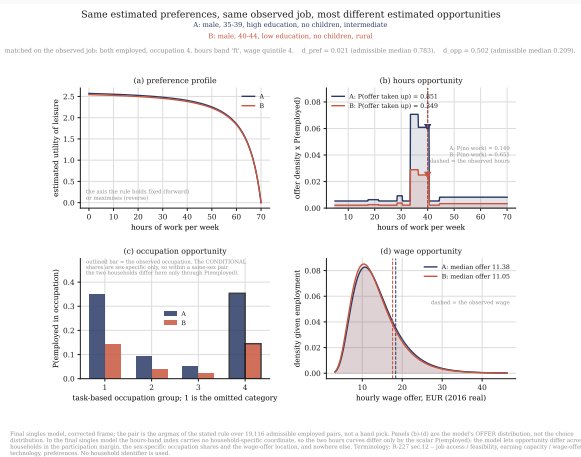
Then draw the line the evidence supports and no further: **the validation claim is limited to the hours-distribution feature for which the model and the external source have commensurable objects — in particular the statutory-band concentration.**

The EU-LFS wish-more-hours and desired-minus-actual-hours profiles, panels (a) and (c), are **separate descriptive context**. Because the model has no desired-hours or underemployment counterpart, they are not interpreted as model predictions or as validation of the estimated opportunity weights, and no association between them and any estimated quantity is computed or shown. Report them descriptively if asked — 20.3 per cent of employed people in the priced bands report wanting more hours, declining with hours worked (23.7 per cent at the statutory band, 7.6 at long hours) — and stop there.

Two things not to skip. The external population is all adults aged 20–60 while the sample is single-adult households, so **a gap here is a population difference, not a prediction error**. And the matched employer–employee earnings source is simply not available in the accessible data, so no external wage comparison is attempted and none is faked.

25-MIN: compress to 1.0 — panel (b) and the population caveat; drop panel (a).

The welfare measure, and two real households



same preference profile (4.61 and 4.56) · employment probability **0.851** against **0.349**

[1.5 min] Define the measure in one sentence and say what normative choice it encodes: what flat consumption level, available at every job in this household's **ex-ante estimated opportunity distribution**, would leave them as well off as they are — so pay is compensated and the set is the household's own responsibility.

Say "estimated opportunity distribution", not "the household's own set": the empirical object is a density the model estimates, not an observed menu, and the welfare integral is taken over it ex ante rather than over realised draws.

Then the pair, and stress that it was **selected by a rule fixed before any household was looked at**, from 19,116 admissible pairs matched on observed employment, occupation, hours band and wage quintile: two single men, both working 40 hours in the same occupation in the same wage quintile, whom the model gives essentially the same preference profile — leisure weights 4.61 and 4.56 — and very different opportunity: **employment probability 0.851 against 0.349**.

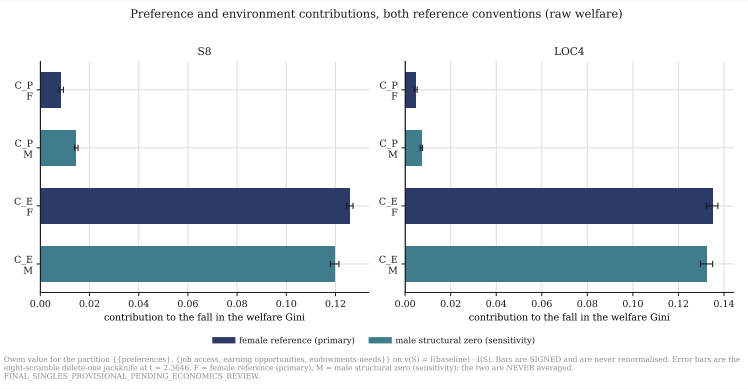
Point at where the difference is *not*: their median wage offers are €11.38 and €11.05, within thirty cents. The whole gap is on the participation margin, which is where region, urbanisation and the local unemployment rate enter — so this pair is the household-level face of the geographic split on slide 17.

This is slide 3's panel (a) on real estimates. Repeat once that these panels are **densities, not job draws**.

If asked how this pair was chosen, or why it differs from an earlier version: a defect in the selection code left the ranking's opportunity distance without the region term. It is repaired, the ranking was re-run over the same admissible pairs under the same rule, and this is what it returns — a cleaner pair, separated by $2.40\times$ rather than $2.08\times$, whose gap is geographic rather than in the wage offer. **No number in the welfare decomposition moved**; the defect was confined to this illustration.

25-MIN: KEEP (1.5) — the measure and the pair, no panel walk-through.

The headline



The environment $93.70 \pm 0.61\%$ · Preferences $6.30 \pm 0.61\%$

[2.5 min] Read the sentence as it is written, because it is generated from the table with a guard and its qualifiers are not decoration: *within the currently modelled non-preference environment, budget-side endowments and needs are the largest nested contribution under both positive models and both reference-preference conventions.*

Baseline inequality is a Gini of 0.134; equalising the environment alone takes it to 0.031 (equalising preferences alone *raises* it, to 0.148316); equalising both takes it to numerical zero — and that last point is a **tested property, not an assumption**: the fully common state is zero to machine precision, which is what licenses reporting shares at all.

Then the licensed reading, which must be said out loud: **no channel “explains” a percentage**. A contribution is the value of an equalisation operator in a cooperative game — the form is *equalising this channel removes this share of baseline inequality* — not a variance share and not a causal effect.

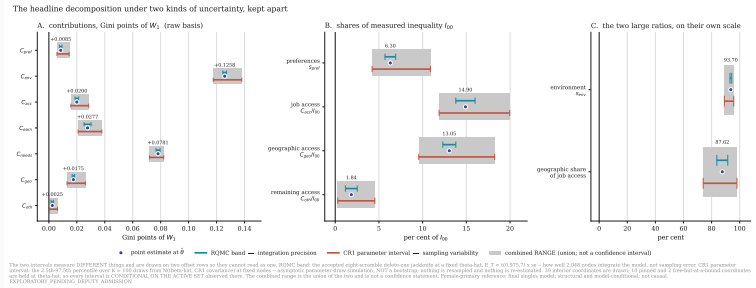
CLICK 2. Be exact about the bars, because there are now **two kinds** and they must not be read as one. The **integration band** says how well the nodes integrate the model at a fixed parameter vector. The **parameter interval** says how much the estimate would move across samples — propagated from the cluster-robust asymptotic distribution, **not a bootstrap**, nothing resampled and nothing re-estimated. They are drawn on offset rows and are **never merged**; the shaded union is not a confidence interval.

The parameter interval is the wider by 3 to 8 times, so the preference share is 6.30 per cent with an interval of **[4.22, 10.89]** and the environment share 93.70 with **[89.11, 95.78]** — the environment stays dominant across it, and the sign of the preference contribution does not change within it. Every parameter interval is **conditional on the active set** at the estimated parameter vector, over 100 draws from the asymptotic sampling distribution.

One counter-intuitive fact worth 30 seconds if the room is engaged: equalising preferences *raises* measured inequality, because the households with the worst opportunity sets are the ones whose own tastes currently do most to reconcile them to those sets.

25-MIN: KEEP (2.5) — this is the slide the talk exists for.

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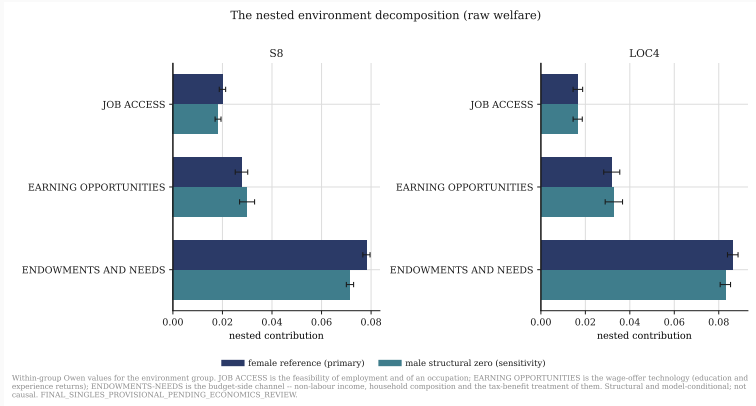
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The environment $93.70 \pm 0.61\%$ · Preferences $6.30 \pm 0.61\%$

parameter interval: preferences [4.22, 10.89] the environment [89.11, 95.78]

Inside the environment



Household endowments and needs $58.18 \pm 1.47\%$ · Earning opportunities $20.62 \pm 1.63\%$ · Job access $14.90 \pm 1.10\%$
market side together: $35.51 \pm 0.95\%$

[1.5 min] Endowments and needs 58.18 per cent (plus or minus 1.47), earning opportunities 20.62 (1.63), job access 14.90 (1.10) — so the two market-side channels together are **35.51 per cent (0.95)**, and that combined figure is reported separately from endowments and needs on purpose.

Two disciplines. First, **do not say job opportunities explain the environment share**: the environment is the *complete* non-preference environment, and the market-side part of it is a third, not all of it.

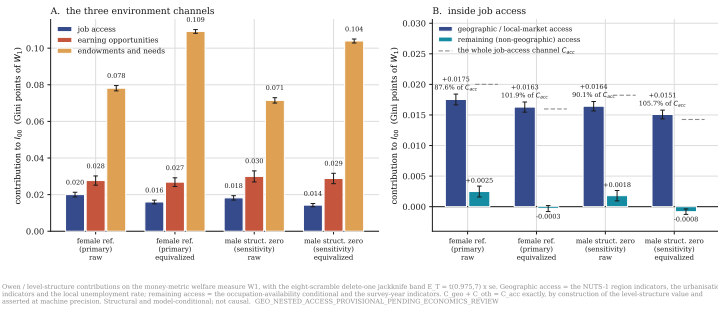
Second, **no ordering is claimed between job access and earning opportunities** — the point estimates order them and are printed, but the access margin is exactly the one that does not separate capability from availability, so an ordering claim would be a claim about something the design does not identify.

If asked why the budget channel is so large, give the honest reading: this is France 2016, a system with substantial means-tested transfers, and the result is partly a statement about that system. It is not portable without re-estimation.

25-MIN: KEEP (1.5) — the three numbers, the combined market-side figure, and the no-ordering rule.

Inside job access: it is almost all geography

The nested geographic split of the job-access channel (final singles model, money-metric W_1)



geographic access $13.05 \pm 0.74\%$ raw, $9.87 \pm 0.55\%$ equivalized — 87.62% and 101.92% of job access [74, 98]

This is not a finding that region drives welfare inequality.

[2.0 min] Two numbers and one caveat, and the caveat is the point.

Equalising the geographic access environment removes **13.05 per cent (0.74)** of baseline inequality on the raw basis and 9.87 (0.55) equivalized, against a whole job-access channel of 14.90 and 9.68. So geography is **87.62 per cent (3.89)** of the channel raw and 101.92 (3.11) equivalized.

Never say that ratio without its parameter interval — **[74, 98] per cent on the raw basis**. It is a ratio of two contributions one of which is small, so its sampling distribution is much wider than its integration band suggests.

Say why a share can exceed one before anyone asks: shares here are **signed and never renormalised**, so the residual is negative on the equivalized rows, not missing.

Then the caveat, delivered as a limitation of the specification rather than as a result. The two sub-players do not have comparable support. The geographic block varies household by household over eight regions, three urbanisation categories and a region-education-sex unemployment rate. The remaining access block is the occupation-availability conditional, and this specification conditions it on **sex alone** — a two-cell table — plus year indicators that are identically zero on a one-year sample. Almost all the between-household dispersion the access operator *can* remove is geographic **by construction**. The honest reading is about the composition of the access block, not about the importance of place.

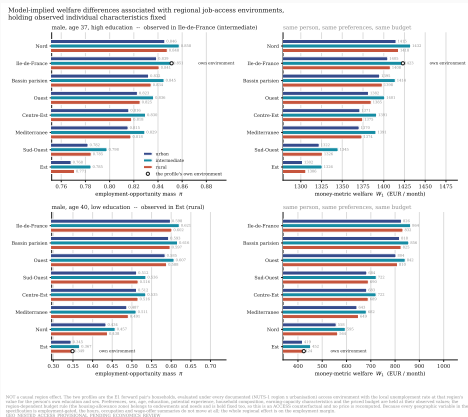
And the sentence that stops the obvious misreading: **this is not a finding that region drives welfare inequality**. Job access as a whole is 14.90 per cent of measured inequality raw and 9.68 equivalized, against 58.18 and 66.10 for endowments and needs. If asked about regional policy: the only regional dependence in the tax-benefit system here is the housing-allowance zone, and that belongs to endowments and needs, deliberately outside this split.

CLICK 2 — optional, and the first thing to drop. This takes the two matched households of slide 14 and moves them across all 24 region × urbanisation environments, holding preferences, demographics, earning-capacity characteristics, the priced budget and the region-dependent budget rule fixed — an access counterfactual with no price recomputed. The high-education man's employment-opportunity mass runs 0.768 to 0.858 and his welfare €1,302 to €1,432; the low-education man's, **0.345 to 0.621** and **€419 to €864**. Handle that second range carefully if you show it: it is not that moving would double his welfare, it is that a household already at a low participation probability sits on the steep part of the map from access index to employment margin, and his own environment is the least favourable of the twenty-four.

Because every geographic variable here enters only when the household works, the whole regional difference is **on the employment margin**. Say the label out loud — **this is not a causal region effect**; nothing here identifies what would happen to someone who moved.

25-MIN: CUT to backup.

Inside job access: it is almost all geography



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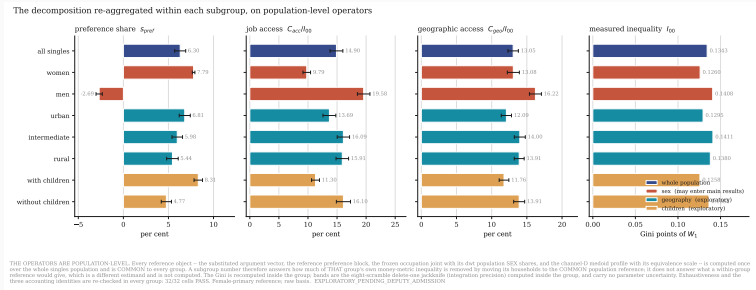
And the sentence that stops the obvious misreading: **this is not a finding that region drives welfare inequality**. Job access as a whole is 14.90 per cent of measured inequality raw and 9.68 equalized, against 58.18 and 66.10 for endowments and needs. If asked about regional policy: the only regional dependence in the tax-benefit system here is the housing-allowance zone, and that belongs to endowments and needs, deliberately outside this split.

CLICK 2 — optional, and the first thing to drop. This takes the two matched households of slide 14 and moves them across all 24 region × urbanisation environments, holding preferences, demographics, earning-capacity characteristics, the priced budget and the region-dependent budget rule fixed — an access counterfactual with no price recomputed. The high-education man's employment-opportunity mass runs 0.768 to 0.858 and his welfare €1,302 to €1,432; the low-education man's, **0.345 to 0.621** and **€419 to €864**. Handle that second range carefully if you show it: it is not that moving would double his welfare, it is that a household already at a low participation probability sits on the steep part of the map from access index to employment margin, and his own environment is the least favourable of the twenty-four.

Because every geographic variable here enters only when the household works, the whole regional difference is **on the employment margin**. Say the label out loud — **this is not a causal region effect**; nothing here identifies what would happen to someone who moved.

25-MIN: CUT to backup.

Job access does more of the work among men



[1.5 min] Say what the estimand is before the number, because it is easy to hear as something else. Every reference object is computed **once over the whole population and is common to every group**, so a bar answers *how much of this group's own inequality is removed by moving its households to the common population reference*. It is a **within-subgroup** decomposition, not a contribution of the characteristic: nothing here splits inequality into between-sex and within-sex parts, and nothing here is causal.

The finding: **job access carries 19.58 per cent of men's measured inequality against 9.79 of women's** — and it holds under both reference conventions and on both bases, 18.44 against 9.02 under the male reference, 16.74 against 7.17 equivalized. The geographic part is larger for men too, 16.22 against 13.08.

Say carefully what it does not mean: it is not that men face worse access, because the decomposition is not a level comparison. It is that the dispersion of access within men does more of the work among men than the dispersion of access within women does among women.

Then the disclosure, and do not skip it. The men's preference share is -2.69 per cent under the female reference and +2.78 under the male one. That is a **sign change**, not a magnitude sensitivity, and **no statement about the sign of the male preference contribution is licensed** — so make none. It is worth pausing on: the population preference share is positive under both conventions, so the headline's sign stability does not extend to every subgroup of it. This is the sharpest demonstration in the talk of how much rides on a normative choice.

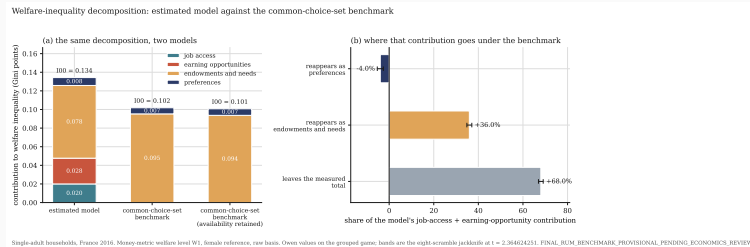
If asked about other splits: urbanisation and children were computed on the same construction, neither carries a finding that survives the same scrutiny, and both are in the appendix.

25-MIN: CUT to backup.

job access: $19.58 \pm 1.08\%$ of men's measured inequality against $9.79 \pm 0.65\%$ of women's

Men's preferences share: -2.69% and +2.78% across the two reference conventions — a sign change. No sign statement is licensed.

What a conventional model would have found



preferences share 6.4% against 6.3% · measured inequality -24.2% raw · sex
leisure gap +0.428 → -1.991

[2.5 min] This is the paper's own counterfactual about method and it is worth doing properly, because **the result is not the one the premise predicts** and saying so is what makes the rest credible.

Set it up in one line: every household-indexed argument of the availability object is replaced by a common value, so the benchmark is a conventional model with hours constants, a fixed cost of work and a sex-specific leisure block — what a careful analyst without an opportunity object would have fitted.

Then three findings, in this order. **First, fit does not separate them.** The benchmark is 129.1 nats worse on the objective (18151.85 against 18022.76 over 41 free parameters; a likelihood-ratio statistic of 258.2 on 25 degrees of freedom, quoted as an upper bound because the benchmark is not nested), but its marginal fit is *as good or better* — hours 0.0080 against 0.0083, occupation 0.0029 against 0.0034, wage quintiles 0.0203 against 0.0261. A referee judging on marginal fit alone would prefer the model with no opportunity object in it. That is the argument for not judging on marginal fit alone.

Second, the availability information goes into the taste parameters. The 5 hours-band constants reappear in the benchmark's *utility* index at essentially their availability values — mean absolute difference 0.037. The same constants, read twice. And the male-minus-female leisure gap **reverses sign**, from +0.428 to -1.991: measured leisure preference shifts toward the group facing the worse offer environment, exactly as the confound predicts.

Third — and this is the honest surprise — the preference share of the welfare decomposition barely moves: 6.4 per cent under the benchmark against 6.3 under the preferred model. The omitted market-side share does not become taste. It mostly leaves the total — measured inequality falls -24.2 per cent on the raw basis and -15.3 equalized — and what stays is relabelled onto endowments and needs.

Say the licensed sentence: *behavioural coefficients can absorb omitted availability structure; omitting heterogeneous opportunities materially lowers measured money-metric inequality; the welfare-decomposition preference share changes little; and the missing opportunity contribution is reallocated primarily within the non-preference environment rather than into preferences.*

Do **not** claim it demonstrates a large welfare-level preference-misattribution effect.

If asked why that matters for the paper's contribution: the case for an opportunity object is not that it rescues the decomposition from attributing everything to taste — on this measure it does not. It is that without one the estimated preferences point the wrong way, the level of inequality is understated by a fifth to a quarter, and the environment's composition is misdescribed.

25-MIN: compress to 1.5 — the reversed sex gap and the preference-share result; drop the fit comparison.

What is fragile, and what is not

perturbation	preferences share	the environment side
reference-preference convention	6.30 → 10.89%	< 3.5%
a bound on a preference curvature	≈ 13% move	< 3.5%
pinning the couples male leisure block	up to 73% move	< 3.5%

No sign and no ordering changes anywhere. Market-side total 35.51% against 35.88% across references.

[2.0 min] This is the most useful slide for a critical audience and it is a strength, not a concession.

Be precise about which perturbation produces which range — they are different objects and it is easy to conflate them.

Changing only the reference-preference convention on the preferred singles model moves the preference contribution from **6.30 to 10.89 per cent on the raw basis** and from **7.98 to 10.97 per cent on the coalition-consistent equivalized basis**. These paired values are reported as ranges and never averaged.

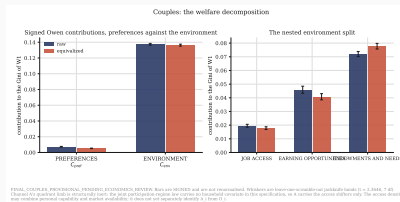
The broader **3.3–11.0 per cent** envelope is across all eight rows of the exact table and therefore also reflects the positive-model and equivalization choices; **it is not a reference-only range**.

The other two perturbations: the age-bound diagnostic moves the preference share by about 13 per cent, and pinning the couples male leisure block moves it by up to 73 per cent. The environment side moves by under 3.5 per cent in all of them, and no sign and no ordering changes anywhere.

The market-side total is nearly unchanged across reference conventions — 35.51 against 35.88 per cent on the raw basis — so reference sensitivity is concentrated in the preference/environment split rather than in the environment's internal structure.

25-MIN: KEEP (1.5) — the three rows and the range rule; absorb two sentences of slide 22.

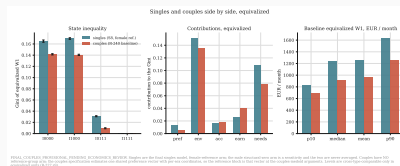
Couples



the couples decomposition

2,275 couples · same nested order, same the environment dominance

The couples preferences share is not structurally robust. Singles remain the quantitative headline.



singles and couples, equivalized

[2.0 min] 2,275 opposite-sex couples, one specification and no search.

Lead with the finding that was not targeted and is the most striking piece of internal consistency in the paper: the **female statutory-week peak in couples reproduces the singles value to within three hundredths** — 2.5474 against 2.5795, a third of a standard error — on a different frame, a different choice set and a joint likelihood.

Then the substance: same nested order on both bases, same environment dominance.

Then the discipline, and do not sorten it. Synthetic-recovery diagnostics identified the male leisure block as the weakly recovered part of the couples specification *before* the welfare sensitivity exercise was run, and that exercise confirms it bites: the preference contribution moves by up to 73 per cent, so **no single couples preference percentage is presented as structurally robust**, and every couples preference figure travels with its range.

One directional difference from singles, stated because it is real: for couples, equalising preferences *reduces* measured inequality, the opposite of the singles result, so the singles interpretation does not transfer.

If asked about pooling: pooled equivalized inequality is well defined and reported, but the pooled *decomposition* is not exhaustive — the two household types have different reference bundles, so the fully common pooled state is 39.4 per cent of the pooled baseline. That is an **open welfare-reference question, not a numerical failure**.

25-MIN: compress to 1.0 — the statutory-week replication, the same order, and the sensitivity caveat.

Limits, and what would identify more

limit	what would change it
access versus capability	external job-offer or access-rate moments
persistent heterogeneity	repeated choices per household
no desired-hours target	external desired-hours moments
one country, one year, one system	re-estimation; no portability claimed

Multi-adult households are excluded. Welfare magnitudes are provisional pending one bounded economics review.

[1.5 min] Four limits, thirty seconds each at most.

Access versus capability — separating them needs external moments on job-offer or access rates.

Persistent heterogeneity — needs repeated choices per household, which would supply the second contrast slide 9 showed is missing.

The hours block has no desired-hours target — external desired-hours moments would discipline it against a stated target rather than inferring both from realised hours.

One country, one year, one system — nothing here is portable without re-estimation, and no portability is claimed.

Say the honest scope line: multi-adult households are excluded, and their admission needs five assumptions this design leaves unstated. Also state, once, that the welfare magnitudes are **provisional pending one bounded economics review**.

25-MIN: CUT — fold two sentences into slide 20 and slide 23.

- **The method.** A household-specific opportunity distribution estimated jointly with **preferences**, every job priced through the real tax–benefit system.

[1.0 min] Close on the contribution rather than the number.

The method: a household-specific opportunity distribution estimated jointly with preferences, with every job priced through the real tax–benefit system, and a decomposition that is exhaustive **by measurement rather than by assumption** — the fully common state is numerically zero.

The finding: the non-preference environment carries roughly nine-tenths of measured money-metric welfare inequality (93.70 per cent), and inside it the budget side carries more than the market side (58.18 against 35.51).

The margin: the preference share is the fragile quantity, it is reported as a range across the two normative conventions (6.30 to 10.89 per cent), and it is where the next round of evidence should go.

Then stop talking.

25-MIN: KEEP (1.0) — absorb the two limits sentences.

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- **The finding.** The environment carries 93.70% of measured money-metric welfare inequality; inside it the budget side carries more than the market side.
- **The margin.** The **preferences** share is the fragile quantity: 6.30% to 10.89% across the two normative conventions.

[1.0 min] Close on the contribution rather than the number.

The method: a household-specific opportunity distribution estimated jointly with preferences, with every job priced through the real tax–benefit system, and a decomposition that is exhaustive **by measurement rather than by assumption** — the fully common state is numerically zero.

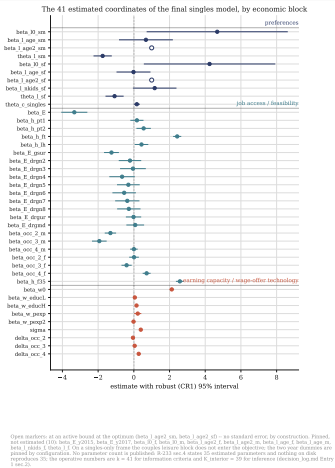
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Then stop talking.

25-MIN: KEEP (1.0) — absorb the two limits sentences.

B1 — The estimated coefficients



41 estimated structural parameters, robust 95% intervals

Held for: “What are the actual parameter estimates?”

The final singles model carries **41 estimated structural parameters**. The full table with standard errors is in the paper’s Appendix A.

Read off the two things worth reading: the leisure curvatures differ sharply by sex and are precisely estimated, and the occupation-availability weights differ strongly by sex — men concentrated in and around routine manual, women’s availability tilted toward nonroutine cognitive. That sex-specific structure is what keeps occupational sorting out of the taste block.

Note that two coordinates rest on an active bound and carry no standard error, and that the ten pinned coordinates are not displayed.

B2 — Exhaustiveness: how the decomposition was made to close

defect	what fixed it
a missing budget channel	added as a fourth player from its own marginals
asymmetric money-metric inversion	reference side moves with the coalition
an incomplete male reference block	structural zero on the child leisure term

Exhaustiveness here is a tested property, pre-stated before any cell was read.

the fully common state is 0.000000 — numerically zero

Held for: “*Why should I believe the four channels are exhaustive?*” — and it is the best question in the room.

This is a methodological result and it should be told as one, because the honest answer is that it did **not** close at first. An earlier implementation failed the exhaustiveness test on both limbs in every arm — equalising everything left an index **three times** the one it started from — and the headline was correctly **halted**.

Three things were found and removed. A missing budget channel, added as a fourth player formed from its own marginals rather than as a relabelled residual. An asymmetry in the money-metric inversion, where the reference side was frozen at the baseline while the attained side moved with the coalition; correcting it removed about 95 per cent of the residual. And a male reference block that was incomplete, because the specification carries a children-in-leisure term for women and none for men.

With all three fixed the fully common state is numerically zero. **Exhaustiveness here is a tested property, and the test was pre-stated before any cell was read.**

B3 — Equivalization, and why the scale has to move

	raw	equivalized
preferences	6.30%	7.98%
the environment	93.70%	92.02%
job access	14.90%	9.68%
earning opportunities	20.62%	16.24%
household endowments and needs	58.18%	66.10%
market side	35.51%	25.92%

Freezing each household's own scale leaves a residual of 0.071799 — precisely the weighted Gini of one over the scale.

Held for: “You divide by an equivalence scale — doesn’t that break the decomposition?”

It does, under the natural convention, and exactly: freezing each household's own scale in all states leaves a residual of 0.071799, which is **precisely the weighted Gini of one over the scale** — the same number to six decimals. Dividing a common numerator by a household-specific divisor reintroduces exactly the dispersion of the divisor.

The fix is the one the inversion already legislates: household composition is an endowments-and-needs object, so the scale moves with the channel that equalises composition, and exhaustiveness is restored exactly.

Equivalizing raises measured inequality by about a fifth and changes no qualitative conclusion, but it does move the market-side total from 35.51 to 25.92 per cent — both are reported and neither is “the” answer.

Then the normative point, and state the justification as **factor consistency, not closure**: the scale moves because household composition is itself part of the endowments-and-needs factor being equalised — the equivalence scale is a function of that composition and is therefore recomputed at the coalition-specific needs profile. **Exact closure is a consequence of this definition, not its justification.**

If instead the normative object were to equalise resources while leaving needs fixed, the household's own scale would remain and the residual would be a genuine needs component requiring different accounting. That alternative is coherent; this paper adopts the first object.

B4 — Why the hours-wish evidence is context, not a moment

ground	what it shows
definitional	the candidate’s optimum tracks pay: 98.1% pay more, median 2.58×
identification	the hours block moves the moment by exactly 0
numerical	24.3% against 20.3%; the profile does not match

Steelman (own ±5% wage neighbourhood): 19.4% — and the hours block is exactly as inert.

Held for: “You have external data on who wants more hours — why not use it to discipline the hours block?”

The honest answer is that it was tried and there is no counterpart to compare against. Three independent grounds.

Definitional: the survey asks about more hours *at the current wage*, and the frame has no such menu — exactly one node per household carries the observed wage, the chosen one, so the candidate's optimum tracks pay, paying more for 98.1 per cent of workers at a median 2.58× the observed wage.

Identification, and this is the decisive one: perturbing any of the six hours-opportunity coefficients moves the moment by **exactly zero** and moves not one household's optimum — removing the opportunity weights removes the very block the moment was meant to discipline.

Numerical: the level can be made to agree and the profile cannot; 24.3 per cent against 20.3, but the external share falls monotonically in hours and the model's does not, and the model puts none at long hours against 7.6 per cent.

The steelman was tried: restricting to the household's own ±5 per cent wage neighbourhood fixes the pay artefact and brings the aggregate to 19.4 per cent — closer than the model deserves — and the hours block is *exactly as inert*.

The fork is architectural: a definition that removes the opportunity weights cannot identify the hours block, and one that retains them is the model's own choice distribution, not an unconstrained-desire object. So the external moment stays descriptive, and no calibration is proposed.

B5 — Where the welfare family comes from

Compensation

unequal productivity should not
become unequal well-being

Responsibility

preference-based choices should not
be compensated

$$\nexists W : \text{Full Compensation} \wedge \text{Full Responsibility}$$

[Fleurbaey & Maniquet, SCW 2005]

$W^1(z, R, A; \mathbf{y}^w)$ — equal pay on the feasible set;
the measure this paper takes to the data

Haydar and Maniquet, *Jobs and Well-Being Measurement* (companion paper).

Held for: “Where does this welfare measure come from, and why this one?”

Reused from the companion theory talk; cited, not merged with slide 14.

Two principles we would like to respect. **Compensation:** people should not be penalized for unequal productive circumstances.
Responsibility: people should be held responsible for their preferences, especially their tastes for work and leisure.

They are attractive separately but cannot both be imposed in full: no well-being measure satisfies Full Compensation and Full Responsibility together. So the literature is structured around compromises.

W^1 is one of them, and it is the one this paper takes to the data. Ignore the actual pay profile; put the same consumption level on all jobs in the individual's ability set; among these feasible jobs the preference relation selects one; move the common level until being at that selected job is exactly indifferent to the actual bundle.

It satisfies Independence of $\mathbf{y}$ — it ignores the actual pay profile — and Responsibility for Equal Pay — once all feasible jobs pay the same, the choice among them reflects preferences over jobs.

The empirical translation on slide 14 replaces the ability set A with the *estimated opportunity density*, which is where this paper's work is.